Instrumental Variables: The Lottery Example

Experimentation, A/B Testing and Causal Inference (SDA-DSC-213), SDAIA Academy. Handout for Day 3, Hour 4 (slides 109 to 112). All five Day 3 handouts use the same story: a tutoring programme at one school.

Step 1. The story

Tutoring slots are limited, so the school runs a **lottery**: 100 students apply, 50 win a slot. Winning does not force anyone to attend, and losers can sometimes find tutoring on their own. We want the effect of **attending tutoring** on the exam score, but attendance is chosen, so it is confounded. The lottery result is not.

Symbol	Meaning	Here
Z	the instrument	won the lottery (1) or not (0)
T	the treatment	attended tutoring (1) or not (0)
Y	the outcome	exam score

Step 2. The three conditions (slide 110)

Condition	Meaning	Here
Relevance	Z changes T	winners attend far more often; testable
Exclusion	Z affects Y only through T	a lottery ticket cannot raise a score by itself; an assumption
Independence	Z is as good as random	it is a lottery; true by design

Step 3. The data

Group	Students	Attended tutoring	Attendance rate	Mean score
Winners (Z = 1)	50	40	0.80	70
Losers (Z = 0)	50	10	0.20	64

Step 4. Three numbers

Intention to treat estimand: ITT: the effect of winning, regardless of attendance.

```
ITT = mean(Y | Z=1) − mean(Y | Z=0) = 70 − 64 = +6
```

First stage: how much the lottery moved attendance.

```
first stage = P(T=1 | Z=1) − P(T=1 | Z=0) = 0.80 − 0.20 = 0.60
```

The Wald ratio estimator: Wald = ITT / first stage, the simplest form of two-stage least squares:

```
τ̂_IV = 6 / 0.60 = +10
```

Reading: winning raised scores by 6 points on average, but winning only changed attendance for 60 percent of students. Spread the 6 points over that 60 percent and the effect of attending, for those the lottery moved, is 10 points estimate: +10.

Step 5. Who is the 10 points about: LATE (slide 111)

Type	Behaviour	Share here	IV says about them
Compliers	attend if they win, not if they lose	0.80 − 0.20 = 0.60	the 10 points is their effect
Always-takers	attend either way	0.20	nothing
Never-takers	never attend	0.20	nothing
Defiers	attend only if they lose	assumed 0 (monotonicity)	ruled out

So the estimand IV recovers is the **local** average treatment effect estimand: LATE, the effect for compliers. It is not the ATE. A different instrument would move different students and could give a different, equally valid, LATE.

Step 6. Why not just compare attenders with non-attenders

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naive = mean(Y | T=1) − mean(Y | T=0) (mixes lottery winners, self-starters and the rest)
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Attenders include always-takers, the most motivated students, who would score high anyway. The naive gap carries that motivation; the Wald ratio does not, because it only uses variation created by the lottery.

Step 7. Weak instruments (slide 112)

If the lottery barely changed attendance, say 0.45 versus 0.40, the first stage would be 0.05 and the Wald ratio would divide a noisy ITT by a tiny number. Report the first-stage strength: the rule of thumb is a first-stage F statistic above 10. Below that, the IV estimate leans back toward the biased naive comparison and its interval is unreliable.

Three things to remember

1. IV = ITT divided by the first stage. Both parts must be reported.

2. The result is the effect on compliers, not on everyone.

3. Exclusion cannot be tested; argue it from how the instrument works.